Chiral relations for entanglement in quantum self-phase modulation

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We investigate chiral selection rules for bipartite and multipartite entanglement in the radial mode structure generated in self phase modulation with orbital angular momentum (OAM). While OAM conservation is a trivial selection rule for entanglement, a less intuitive rule applies to the radial structure. Depending on the relative chirality of the interacting modes, entanglement can be concentrated between two radial modes or distributed in a multiparty configuration determined by the OAM values involved. For the full pure field, the first-order correlations certify genuine multipartite entanglement when the remaining modes are grouped with one selected mode. Finite-distance Bogoliubov calculations and optimized witnesses further test entanglement of the reduced selected modes. This opens the possibility of generating and controlling multipartite entanglement in the radial structure.

I. INTRODUCTION

The quest for quantum devices based on optical resources is an intense and active field of research, which has already delivered important tools for quantum communication, computing and metrology. This quest has increased considerably by the possibility of shaping light in its different degrees of freedom, such as time envelope, spectral and spatial modes [1, 2]. Both discrete and continuous variable quantum information processing can benefit from these developments [3–5]. Nonlinear optical processes are the common tool for generating entangled quantum states, as it allows the energy transfer between different optical modes, constrained by different conditions such as energy conservation and phase matching, for example [6–10]. These conditions are the basis for entanglement, as they impose different kinds of quantum correlations among the modes interacting in the nonlinear process [11–14]. Energy and momentum conservation are behind the generation of time-frequency and position-momentum entanglement, respectively. These conditions rule nonlinear optical processes such as three- and four-wave mixing, second harmonic generation and self-phase modulation[15–25].

The possibility of shaping the phase and polarization distributions with spatial light modulators and q-plates has become a powerful tool for controlled generation of customized entangled states both in linear and nonlinear optics. These developments brought the orbital angular momentum (OAM) of light to the toolbox of quantum information technology. Its role in nonlinear processes has also been unveiled by a plethora of scientific papers devoted to the constraints imposed by OAM and vortex formation in nonlinear wave mixing. In polariton superfluids, the interplay between phase, spin and flow gives rise to a rich variety of spatial structures, including quantized vortices, spin textures and vortex flows [26–30]. A recent work has further demonstrated the spectroscopy of collective excitations associated with stationary, multiply quantized vortices, opening new possibilities for exploring vortex dynamics in driven-dissipative quantum fluids of light [30].

A less exploited degree of freedom involves the radial structure of paraxial modes [8, 15, 16, 31–33]. The Laguerre-Gaussian (LG) solutions of the paraxial wave equation are the natural basis for describing OAM and the radial structure, which are the suitable labels for their identification. These modes are characterized by two integer numbers, namely its topological charge and radial index. The topological charge corresponds to the azimuthal variation of the beam phase around the propagation direction and the radial index determines the number of intensity rings. While the OAM conservation selection rule has been demonstrated in different quantum optical processes, the constraints related to the radial structure still deserves careful investigation.

Self-phase modulation is a particularly interesting phenomenon, where the intensity of an optical beam alters the effective refraction index of a nonlinear medium, thus affecting the beam's phase evolution along propagation through the medium [34]. This effect can also be observed in Bose-Einstein condensates and atomic vapors [35–37]. In a previous work we investigated the formation of a radial structure in self-phase modulation driven by an OAM beam and determined the radial spectrum generated in terms of the topological charge of the input beam. This established an interesting radial-angular coupling promoted by the radial selection rules imposed on the nonlinear process [38].

In the present work we investigate the consequence of radial-angular coupling and the associated selection rule to entanglement generation. In particular, we will discuss bipartite and multipartite entanglement generated in the process, which can serve as a possible platform for quantum information protocols based on these resources. This platform is particularly interesting due to the wide range of experimental tools for its implementation, going from polariton superfluids to atomic vapors.

Section III derives the first-order correlations from the full fluctuation equation. Section IV establishes their radial selection rule and the resulting pairwise, multimode bipartite, and grouped pure-state entanglement. Section V then uses the diffraction-free Bogoliubov solution and optimized witnesses to investigate finite propagation distances and the reduced selected modes.

II. SPATIALLY STRUCTURED BOSONIC FIELD

Before addressing the effective Hamiltonian of an optical field undergoing nonlinear propagation in a Kerr medium, let us establish the formal background allowing for the treatment of quantized spatial modes and their relevant quantum properties, including entanglement. Although the quantum theory of spatial optical modes have been extensively discussed in the literature, we would like to set the relevant concepts and definitions that will be used when dealing with the quantized mode structure of the optical field.

We start by considering the normalized positive frequency component of the quantized electromagnetic field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\begin{aligned}\left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right]&=\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ \left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}(\mathbf{r}_2)\right]&=\left[\mathcal{E}^\dagger(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right]=0.\end{aligned}\tag{1}$$

In the above equation and in what follows, $\mathbf{r}=(x,y)$ is the transverse coordinate. One can introduce ladder operators $a(v),a^\dagger(v)$ associated with the spatial mode v by the expression

$$a(v)=\int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}.\tag{2}$$

We then have

$$\begin{aligned}\left[a(v_1),a^\dagger(v_2)\right]&=(v_1;v_2)\equiv\int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r}\,,\\ \left[a(v_1),a(v_2)\right]&=\left[a^\dagger(v_1),a^\dagger(v_2)\right]=0\,.\end{aligned}\tag{3}$$

Note that this construction allows us to deal with partially overlapping or non normalized modes, reducing to the standard form when $(v_1;v_2)=0$ or $(v_1;v_2)=1$ $(v_1=v_2)$. This feature will be useful when dealing with bi- and multipartite entanglement among the modes coupled in the nonlinear medium.

Given a classical field distribution $\alpha(\mathbf{r})$ normalized in such a way that $|\alpha(\mathbf{r})|^2$ represents the photon number per unit area, we may build a coherent state through the expression

$$|\alpha\rangle=\exp\left[a^\dagger(\alpha)-a(\alpha)\right]|0\rangle\,,\tag{4}$$

where $|0\rangle$ is the vacuum state. This state is an eigenvector of the field operator,

$$\mathcal{E}(\mathbf{r})\,|\alpha\rangle=\alpha(\mathbf{r})\,|\alpha\rangle\,.\tag{5}$$

This is the type of state that will be used as input on the nonlinear medium.

The simplest entanglement criteria for bosonic systems are based on the field quadratures, which may be defined through the formula

$$X(v)=\frac{a(v)+a^\dagger(v)}{\sqrt{2}}\,.\tag{6}$$

Here, v is the corresponding field mode. The complementary quadrature is simply defined by shifting v by a $\pi/2$ phase: $P(v)=X(iv)$.

The covariance function is given by

$$\Gamma(v_1,v_2)=\langle\{\Delta X(v_1),\Delta X(v_2)\}\rangle\,,\tag{7}$$

where $\Delta X(v)=X(v)-\langle X(v)\rangle$ and $\{\bullet,\bullet\}$ denotes the anticommutator. The calculation of this observable is simplified if we express it in terms of the field $\mathcal{E}$. For this, we introduce the normal and anomalous correlations, given respectively by

$$H(\mathbf{r}_1,\mathbf{r}_2)=\left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\mathcal{E}(\mathbf{r}_1)\right\rangle-\left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\right\rangle\langle\mathcal{E}(\mathbf{r}_1)\rangle\,,\tag{8}$$

$$F(\mathbf{r}_1,\mathbf{r}_2)=\langle\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle-\langle\mathcal{E}(\mathbf{r}_1)\rangle\langle\mathcal{E}(\mathbf{r}_2)\rangle\,.\tag{9}$$

We may then write

$$\Gamma(v_1,v_2)=\mathrm{Re}\left[(v_1;v_2)+2f(v_1,v_2)+2h(v_1,v_2)\right],\tag{10}$$

where

$$\begin{aligned}h(v_1,v_2)&=\int v_1^*(\mathbf{r}_2)v_2(\mathbf{r}_1)H(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &=\left\langle a^\dagger(v_2)a(v_1)\right\rangle-\langle a(v_2)\rangle^*\langle a(v_1)\rangle\end{aligned}\tag{11}$$

and

$$\begin{aligned}f(v_1,v_2)&=\int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &=\langle a(v_1)a(v_2)\rangle-\langle a(v_1)\rangle\langle a(v_2)\rangle\end{aligned}\tag{12}$$

We further note the following properties, which simplify the consideration of linear combination of modes:

$$\begin{aligned}f(v_1,v_2+\lambda v_3)&=f(v_1,v_2)+\lambda^* f(v_1,v_3)\\ f(v_1,v_2)&=f(v_2,v_1)\\ h(v_1,v_2+\lambda v_3)&=h(v_1,v_2)+\lambda h(v_1,v_3)\\ h(v_1,v_2)&=h(v_2,v_1)^*\end{aligned}\tag{13}$$

These are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. Single-mode correlations

Let us analyze the covariance matrix $\gamma_{jk}=\langle\{\Delta Z_j,\Delta Z_k\}\rangle$ where $\mathbf{Z}=(X(v),P(v))$. With our quadrature normalization, the vacuum variance is 1/2.

We therefore express quadrature variances in shot-noise units (SNU), defining $V_{\rm SNU}(Q) = 2\left\langle(\Delta Q)^2\right\rangle$, so that the vacuum variance is one and the eigenvalues of γ are directly the quadrature variances in SNU. We observe that $h(v,v) \in \mathbb{R}$, so we get

$$\gamma = (1 + 2h(v,v))\,\mathbb{1}_2 + 2\,\mathrm{Re}\,f(v,v)\sigma_z + 2\,\mathrm{Im}\,f(v,v)\sigma_x\ , \tag{14}$$

where $\mathbb{1}_2$ is the 2×2 identity, and σ_x and σ_z are the corresponding Pauli matrices. Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi}f(v,v)$. Therefore, we can diagonalize γ by choosing $\phi = [\arg f(v,v) \pm \pi]/2$ and obtain the eigenvalues

$$\lambda_\pm = 1 + 2\left[\,h(v,v) \pm |f(v,v)|\,\right]\ . \tag{15}$$

We then see that the quadrature rotated by ϕ has minimum variance λ_- in SNU, while the one rotated by $\phi + \pi/2$ has maximum variance λ_+. Furthermore, we have squeezing if $|f(v,v)| > h(v,v)$.

B. Duan Criterion for two-mode entanglement

Besides the internal correlations characterizing squeezing or excess noise within each individual mode, we wish to describe inter mode correlations and entanglement. Consider two orthonormal states v_1, v_2, i.e., $(v_j, v_k) = \delta_{jk}$. The Duan criterion [39] states that if

$$\begin{aligned} D = &\frac{1}{2}\left\langle\left[\Delta X(v_1) + \Delta X(v_2)\right]^2\right\rangle \\ &+ \frac{1}{2}\left\langle\left[\Delta P(v_1) - \Delta P(v_2)\right]^2\right\rangle < 1\ . \end{aligned} \tag{16}$$

then the states are entangled.

Let us define new modes $v_\pm = (v_1 \pm v_2)/\sqrt{2}$, so that

$$D = \left\langle\Delta X(v_+)^2\right\rangle + \left\langle\Delta P(v_-)^2\right\rangle\ . \tag{17}$$

By applying (10) we obtain

$$\begin{aligned} D &= 1 + h(v_+,v_+) + h(v_-,v_-) \\ &\quad + \mathrm{Re}\left[f(v_+,v_+) - f(v_-,v_-)\right] \\ &= 1 + h(v_1,v_1) + h(v_2,v_2) + 2\,\mathrm{Re}\,f(v_1,v_2)\ . \end{aligned} \tag{18}$$

Once again, we can maximize the possible violation by rotating the quadrature of one of the modes, say, v_1 by an angle $\phi = \arg f(v_1,v_2) \pm \pi$, which gives the optimal value

$$D_{\rm opt} = 1 + h(v_1,v_1) + h(v_2,v_2) - 2|f(v_1,v_2)|\ . \tag{19}$$

III. THE KERR SYSTEM

We now turn to the quantum description of the Kerr dynamics. Let us consider a monochromatic light field propagating through a $\chi^{(3)}$ nonlinear medium, undergoing self-phase modulation. The input field is assumed to be initially prepared in a coherent state of a spatial mode. Besides self-phase modulation, the input mode is coupled to the vacuum fluctuations of other modes, sharing energy and information through the nonlinear interaction. As we will see, this creates a rich dynamics where energy and information exchange is subjected to selection rules that impose a chiral relation that couples the radial and angular degrees of freedom of the interacting modes and determine the formation of entangled clusters.

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z\mathcal{E}(\mathbf{r},z) + \nabla^2\mathcal{E}(\mathbf{r},z) = g\,\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z)\,, \tag{20}$$

where g is the Kerr nonlinear coupling constant and ∇^2 denotes the laplacian in the transverse direction. Here, we employ a short-distance expansion retaining the full fluctuation equation, which will form the basis of the derivation of our selection rules. Later, we introduce and a diffraction-free, linearized fluctuation treatment without a Taylor expansion in distance, which will allow us to check if the selection rules survive for longer propagation distances.

To start our derivation, let $\alpha_c(\mathbf{r},z)$ denote the solution of the classical equations of motion

$$2ik\partial_z\alpha_c + \nabla^2\alpha_c = g\,|\alpha_c|^2\alpha_c \tag{21}$$

under the initial condition $\alpha_c(\mathbf{r},0) = \alpha(\mathbf{r})$, where $|\alpha\rangle$ is the input coherent state. Let us also define the fluctuation field

$$\eta(\mathbf{r},z) = \mathcal{E}(\mathbf{r},z) - \alpha_c(\mathbf{r},z) \tag{22}$$

We observe that our initial state has vacuum like properties with respect to $\eta(\mathbf{r},0)$. For example, it is annihilated by it:

$$\eta(\mathbf{r},0)\,|\alpha\rangle = 0 \tag{23}$$

Furthermore, any normally ordered monomial containing at least one fluctuation operator has zero expectation value at $z = 0$.

The correlation functions, which are our observables of interest, can also be compactly written in terms of η:

$$\begin{aligned} F(\mathbf{r}_1,\mathbf{r}_2;z) &= \langle\eta(\mathbf{r}_1,z)\eta(\mathbf{r}_2,z)\rangle - \langle\eta(\mathbf{r}_1,z)\rangle\langle\eta(\mathbf{r}_2,z)\rangle, \\ H(\mathbf{r}_1,\mathbf{r}_2;z) &= \langle\eta^\dagger(\mathbf{r}_2,z)\eta(\mathbf{r}_1,z)\rangle - \langle\eta(\mathbf{r}_1,z)\rangle\langle\eta^\dagger(\mathbf{r}_2,z)\rangle. \end{aligned} \tag{24}$$

Next, we write the equation of motion

$$2ik\partial_z\eta + \nabla^2\eta = g\left(2|\alpha_c|^2\eta + \alpha_c^2\eta^\dagger + \Omega\right) \tag{25}$$

where the nonlinear fluctuation terms are

$$\Omega = \alpha_c^*\eta^2 + 2\alpha_c\eta^\dagger\eta + \eta^\dagger\eta^2. \tag{26}$$

We start by considering an approximation where only terms up to $\mathcal{O}(z)$ are retained. From definition, we already have $\langle\eta(\mathbf{r},0)\rangle = 0$. Furthermore, as the equation

of motion (25) contains only normally ordered terms, we also get $\partial_z \langle \eta(\mathbf{r},0)\rangle = 0$, such that

$$\langle \eta(\mathbf{r},z)\rangle = \mathcal{O}(z^2), \tag{27}$$

or, equivalently,

$$\langle \mathcal{E}(\mathbf{r},z)\rangle = \alpha_c(\mathbf{r},z) + \mathcal{O}(z^2). \tag{28}$$

Now we turn to the correlation functions. For compactness, we write $\eta_{j,z} = \eta(\mathbf{r}_j, z)$. In order to calculate the normal correlation, we differentiate the two-operator average:

$$\partial_z \langle \eta_2^\dagger \eta_1\rangle\Big|_0 = \langle (\partial_z \eta_2^\dagger)_0 \eta_{0,1}\rangle + \langle \eta_{0,2}^\dagger (\partial_z \eta_1)_0\rangle. \tag{29}$$

Both terms vanish: the first average has a vacuum-annihilating operator on the right, and the second has one on the left.

Turning, to the anomalous case, we have that

$$\begin{aligned}\partial_z \langle \eta_{1,z}\eta_{2,z}\rangle|_0 &= \langle (\partial_z\eta_1)_0\eta_{2,0}\rangle + \langle \eta_{1,0}(\partial_z\eta_2)_0\rangle \\ &= \langle \eta_{1,0}(\partial_z\eta_2)_0\rangle.\end{aligned} \tag{30}$$

Isolating $\partial_z\eta$ in the equations of motion (25) and substituting above, we see that almost all the terms are zero, because they contain η to the right. The only exception is the term proportional to $\eta^\dagger$, which gives

$$\partial_z \langle \eta_1\eta_2\rangle|_0 = -\frac{ig}{2k}\alpha^2(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2). \tag{31}$$

Since the mean-subtraction terms do not contribute at first order, the correlations are

$$\begin{aligned}F(\mathbf{r}_1,\mathbf{r}_2;z) &= -\frac{igz}{2k}\alpha^2(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2) + \mathcal{O}(z^2),\\ H(\mathbf{r}_1,\mathbf{r}_2;z) &= \mathcal{O}(z^2).\end{aligned} \tag{32}$$

Using the spatial overlap notation

$$(v_1,\ldots,v_n;u_1,\ldots,u_n) = \int d^2\mathbf{r}\,\prod_{j=1}^n v_j^*(\mathbf{r})u_j(\mathbf{r}), \tag{33}$$

projection onto the modes v_1 and v_2 gives

$$\begin{aligned}f(v_1,v_2) &= -\frac{igz}{2k}(v_1,v_2;\alpha,\alpha) + \mathcal{O}(z^2),\\ h(v_1,v_2) &= \mathcal{O}(z^2).\end{aligned} \tag{34}$$

To specify the input and introduce a natural transverse scale, we consider coherent Laguerre-Gaussian modes, with

$$\alpha(\mathbf{r}) = \|\alpha\|\, LG_{pl}(\mathbf{r};w), \tag{35}$$

where the waist-plane modes are

$$LG_{pl}(\mathbf{r};w) = \frac{(-1)^p\mathcal{N}_{pl}}{w}\left(\frac{2r^2}{w^2}\right)^{|l|/2} L_p^{|l|}\left(\frac{2r^2}{w^2}\right) e^{-r^2/w^2 + il\phi}, \tag{36}$$

with radial order p, topological charge l, and normalization constant

$$\mathcal{N}_{pl} = \sqrt{\frac{2}{\pi}}\sqrt{\frac{p!}{(p+|l|)!}}\,. \tag{37}$$

The waist w sets the natural transverse scale. Let us define the Rayleigh range $z_R = kw^2/2$, the nonlinear interaction length $z_{NL} = 4z_R/(g\|\alpha\|^2)$, the dimensionless propagation distance $Z = z/z_{NL} = gz\|\alpha\|^2/(2kw^2)$, and the dimensionless input profile $u(\mathbf{r}) = w\alpha(\mathbf{r},0)/\|\alpha\|$.

Using these definitions, the first-order results take the dimensionless form

$$\begin{aligned}f(v_1,v_2) &\approx -iZ(v_1,v_2;u,u),\\ \Gamma(v_1,v_2) &= \mathrm{Re}(v_1;v_2) + 2Z\,\mathrm{Im}(v_1,v_2;u,u) + \mathcal{O}(Z^2),\\ \lambda_- &= 1 - 2\,|Z(v,v;u,u)| + \mathcal{O}(Z^2),\\ D_{\mathrm{opt}} &= 1 - 2\,|Z(v_1,v_2;u,u)| + \mathcal{O}(Z^2).\end{aligned} \tag{38}$$

For an input angular momentum l_0, the angular integration at first order requires $l_v = l_0$ for squeezing and $l_{v_1} + l_{v_2} = 2l_0$ for entanglement. These OAM conditions are necessary; the corresponding radial overlap must also be nonzero.

IV. SHORT-DISTANCE ENTANGLEMENT STRUCTURE

We now specialize to the LG input $\alpha(\mathbf{r}) = \|\alpha\|\, LG_{0,l_0}(\mathbf{r};w)$ and analyze the selection rules for radial and angular mode coupling.

Our previous work [38] has shown that, in the classical regime, the first order treatment gives a selection rule for the generation of radial orders up to $|l_0|$. The current quantum treatment clarifies that this was, in fact, a selection rule for the mean field value $\langle\mathcal{E}\rangle$ given by (28), whose expression had already appeared on our earlier treatment.

We now derive the radial selection rule for the first-order anomalous correlations and use it to establish pairwise and multimode entanglement at sufficiently small nonzero distances. All detection profiles and their waists are held fixed in this expansion.

A. Radial selection rules and two-mode entanglement

Let $v_2 = LG_{0,l_2}(\mathbf{r};w_2)$ where w_2 is free and l_2 is only constrained by $l_2 \neq l_0$. Now, define $l_1 = 2l_0 - l_2$ and

$1/w_1^2 = 2/w^2 + 1/w_2^2$. We then have that

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2}{w_2}\left(\frac{2r^2}{w_2^2}\right)^{\frac{|l_2|}{2}}\left(\frac{2r^2}{w^2}\right)^{|l_0|} e^{-\frac{r^2}{w_1^2}+il_1\phi}$$
$$= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\, w_1^{2|l_0|+|l_2|}}{w_2^{|l_2|+1}w^{2|l_0|}}\left(\frac{2r^2}{w_1^2}\right)^{\frac{|l_1|}{2}}\left(\frac{2r^2}{w_1^2}\right)^{p_{\max}}$$
$$\times\, e^{-\frac{r^2}{w_1^2}+il_1\phi}, \tag{39}$$

where we have defined

$$p_{\max} = |l_0| + \frac{|l_2| - |2l_0 - l_2|}{2}$$
$$= \begin{cases} 0, & l_0 l_2 \le 0 \\ \min(2|l_0|, |l_2|), & l_0 l_2 > 0 \end{cases}. \tag{40}$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \sum_{p=0}^{p_{\max}} C_p\, LG_{pl_1}(\mathbf{r}; w_1)\,, \tag{41}$$

where

$$C_p = \frac{p_{\max}!\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\, w_1^{2|l_0|+|l_2|+1}}{\mathcal{N}_{pl_1} w_2^{|l_2|+1} w^{2|l_0|}}\binom{p_{\max}+|l_1|}{p_{\max}-p}. \tag{42}$$

Therefore, setting $v_1 = LG_{pl_1}(\mathbf{r}; w_1)$, which is orthogonal to v_2, and considering the two mode subsystem formed by v_1, v_2, we get an optimal Duan violation of

$$D_{\text{opt}} = \begin{cases} 1 - 2C_p|Z| + \mathcal{O}(Z^2), & p \le p_{\max} \\ 1 + \mathcal{O}(Z^2), & p > p_{\max} \end{cases}. \tag{43}$$

For $0 \le p \le p_{\max}$, the coefficients C_p are positive with the chosen LG phase convention. Their linear contribution therefore dominates the $\mathcal{O}(Z^2)$ remainder for sufficiently small nonzero Z, establishing entanglement of each corresponding pair. For $p > p_{\max}$, only the linear coefficient vanishes: Eq. (43) does not decide whether higher-order correlations produce entanglement. The finite radial support is a property of this first-order coefficient in the specified waist basis.

B. Entanglement across multimode bipartitions

Let $\mathcal{A}$ denote the mode v_2, and define the partner parties $\mathcal{B}_p$ as the modes $LG_{p,l_1}(w_1)$ for $p = 0, \ldots, p_{\max}$. These detection modes are mutually orthonormal. Each reduced pair $\mathcal{A}|\mathcal{B}_p$ is entangled at sufficiently small nonzero Z. Because this set is finite, there is a common small-distance interval in which all these pairwise violations hold.

Consider any bipartition $S|T$ placing $\mathcal{A}$ and at least one partner $\mathcal{B}_p$ on opposite sides. If the state were separable across this cut, tracing out all other modes would leave a separable state on $\mathcal{A}|\mathcal{B}_p$, contradicting the pairwise Duan violation. Thus a single entangled pair crossing a cut certifies entanglement across that cut. This conclusion also applies when either side contains additional modes, and requires no purity assumption.

Every nontrivial bipartition of the selected set $\{\mathcal{A}, \mathcal{B}_0, \ldots, \mathcal{B}_{p_{\max}}\}$ separates $\mathcal{A}$ from at least one partner. The reduced state of this set is therefore entangled across every bipartition, or fully inseparable. For mixed states, full inseparability alone does not exclude a convex mixture of states separable across different cuts. A genuine multipartite conclusion requires a further argument.

C. Genuine multipartite entanglement of the full pure state

For the ideal closed, lossless model with a coherent input, the complete field evolves in a pure state. Let $\mathcal{R}$ denote the orthogonal complement of the selected modes, including the pump and all remaining radial and angular modes. For $p_{\max} \ge 1$, consider the $p_{\max} + 2$ parties

$$\mathcal{A} \mid \mathcal{B}_0 \mid \cdots \mid \mathcal{B}_{p_{\max}-1} \mid (\mathcal{B}_{p_{\max}}\mathcal{R}). \tag{44}$$

All degrees of freedom are included, so the state of these grouped parties is pure. Every bipartition separates $\mathcal{A}$ from at least one $\mathcal{B}_p$, possibly contained in the last party, and is therefore entangled by the preceding argument. A pure state is biseparable only if it is separable across at least one bipartition. Consequently, for $p_{\max} \ge 1$, the state is genuinely multipartite entangled with respect to the grouping in Eq. (44). This conclusion holds at any propagation distance for which the complete state remains pure and all the specified reference–partner pairs are entangled; the small-Z restriction applies only to our analytical certification of the pairwise entanglement.

The $p_{\max}+1$ partner modes give a grouping of $p_{\max}+2$ parties at sufficiently small nonzero distance: bipartite when $p_{\max} = 0$, with grouping $\mathcal{A}|(\mathcal{B}_0\mathcal{R})$, and genuinely multipartite when $p_{\max} \ge 1$. The number of guaranteed parties depends on the magnitudes and relative signs of the OAM values through Eq. (40), for the specified detection waists. This is a lower bound on the number of entangled parties in this grouping, not an upper bound on the number of correlated field modes. The complement can instead be assigned to any one of the selected parties without changing the proof.

The purity premise applies to the full modeled field. It does not apply automatically to the reduced selected state $\rho_{\text{sel}} = \text{Tr}_{\mathcal{R}}\,\rho$, which is generally mixed. Tracing out $\mathcal{R}$ can destroy genuine multipartite entanglement, so Eq. (44) does not certify genuine entanglement of the selected single modes alone.

Two questions remain: how far the entanglement structure persists beyond the short-distance regime, and whether genuine multipartite entanglement can be certified after the remaining modes are discarded. We address

these questions through finite-distance covariance calculations and optimized witnesses.

V. ENTANGLEMENT AT FINITE PROPAGATION DISTANCES

A. Diffraction-free Bogoliubov solution

The first-order expressions describe the initial generation of correlations, but extrapolating them to larger distances can give negative values of λ_- or $D_{\rm opt}$. For finite-distance predictions, we return to the full equation for the same decomposition $\mathcal{E} = \alpha_c + \eta$ and adopt a different approximation: we neglect diffraction and discard the nonlinear fluctuation terms Ω in Eq. (25), while retaining the full dependence on propagation distance. This linearization of the fluctuation dynamics is the Bogoliubov approximation, used to describe quantum fluctuations in paraxial propagation through Kerr media [40]. Appendix A solves the diffraction-free Bogoliubov equations without a Taylor expansion in distance. The dimensionless classical field is then

$$u_Z(\mathbf{r}) = u(\mathbf{r}) e^{-iZ|u(\mathbf{r})|^2}, \tag{45}$$

and the mode correlations are

$$\begin{aligned} f(v_1, v_2) = & -iZ(v_1, v_2; u_Z, u_Z) \\ & - Z^2(v_1, v_2, u_Z; u_Z, u_Z, u_Z), \\ h(v_1, v_2) = & Z^2(v_1, u_Z, u_Z; v_2, u_Z, u_Z). \end{aligned} \tag{46}$$

These expressions agree with the preceding first-order result and are exact within the diffraction-free linearized fluctuation equations. Their use at finite distance requires negligible diffraction over the interaction region, which is a reasonable approximation for softly focused beams, as well as a small fluctuation backreaction. The numerical results below use these correlations under those assumptions.

B. Two-mode entanglement

In Fig. 1 we show the violation of the Duan criterion as a function of Z for a few different combinations of l_0 and l_2. In the examples and distance range shown, the modes with $p > p_{\rm max}$ do not violate this criterion. This behavior is consistent with their vanishing first-order coupling, but does not establish an exact finite-distance cutoff or the absence of entanglement.

C. Optimized witnesses and multimode bipartitions

The work published in Ref. [41] introduces a generalized class of entanglement witnesses $\mathcal{Z}$. For a specified separability class, a witness is a real, symmetric, positive semidefinite matrix satisfying $\mathrm{Tr}\,(\mathcal{Z}\gamma_{\rm sep}) \geq 1$ for every covariance matrix $\gamma_{\rm sep}$ of a state in that class. Therefore, if $\mathrm{Tr}\,(\mathcal{Z}\gamma) < 1$, every state with covariance matrix γ lies outside the specified separability class. A value greater than or equal to one is inconclusive and does not establish separability. The Duan criterion is a special case of a witness $\mathcal{Z}$ such that

$$\mathcal{Z} = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \tag{47}$$

The pairwise argument in Sec. IV B establishes entanglement across a cut whenever an entangled pair crosses it. To obtain stronger quantitative certificates and detect correlations missed by the two-mode Duan quantity, we optimize over covariance witnesses. For a fixed bipartition, the FullyWit routine of Ref. [41] minimizes $\mathrm{Tr}(\mathcal{Z}\gamma)$ over witnesses valid for separability across that cut. A value below one certifies entanglement, while a value at or above one is inconclusive.

In Fig. 2 we consider the same setup as in Fig. 1, but use the optimized witness. We do not see a large difference with respect to Fig. 1, which shows that the optimization over the angles used to arrive at (19) already carries us a long way.

Now we turn to more parties. We set $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$. Furthermore, we define the modes as $\mathcal{A} = LG_{0,1}(w_2 = w)$, $\mathcal{B}_0 = LG_{0,3}(w_1)$, $\mathcal{B}_1 = LG_{1,3}(w_1)$ and $\mathcal{C} = LG_{2,3}(w_1)$. Thus $\mathcal{C}$ is a radial mode outside the set coupled by the radial-angular selection rule. The short-distance calculation certifies the pairs $\mathcal{A}|\mathcal{B}_0$ and $\mathcal{A}|\mathcal{B}_1$, while it is inconclusive for $\mathcal{A}|\mathcal{C}$. By Sec. IV B, these two pairs already certify six of the seven bipartitions; the remaining cut is $\mathcal{A}\mathcal{B}_0\mathcal{B}_1|\mathcal{C}$. In Fig. 3 we use the FullyWit routine to find optimized witnesses for all the bipartitions of these four modes. The numerical curves form three groups

- $\mathcal{A}\mathcal{B}_1\mathcal{C}|\mathcal{B}_0$, $\mathcal{A}\mathcal{B}_1|\mathcal{B}_0\mathcal{C}$, $\mathcal{A}\mathcal{C}|\mathcal{B}_0\mathcal{B}_1$ and $\mathcal{A}|\mathcal{B}_0\mathcal{B}_1\mathcal{C}$ show the lowest witness values and hence the largest margins below the separability threshold. These share the fact that $\mathcal{A}$ and $\mathcal{B}_0$ belong to different parties, giving a stronger second-moment entanglement certification across these partitions.

- $\mathcal{A}\mathcal{B}_0\mathcal{C}|\mathcal{B}_1$ and $\mathcal{A}\mathcal{B}_0|\mathcal{B}_1\mathcal{C}$ show intermediate witness values, still maintaining a linear start. Here the entangled pair $\mathcal{A}|\mathcal{B}_1$ crosses the cut, while $\mathcal{A}$ and $\mathcal{B}_0$ belong to the same party.

- $\mathcal{A}\mathcal{B}_0\mathcal{B}_1|\mathcal{C}$ shows the smallest margin below the separability threshold, with a numerical onset consistent with a quadratic fall. Here both first-order entangled pairs lie within $\mathcal{A}\mathcal{B}_0\mathcal{B}_1$, so the individual-pair argument gives no first-order certificate across this cut.

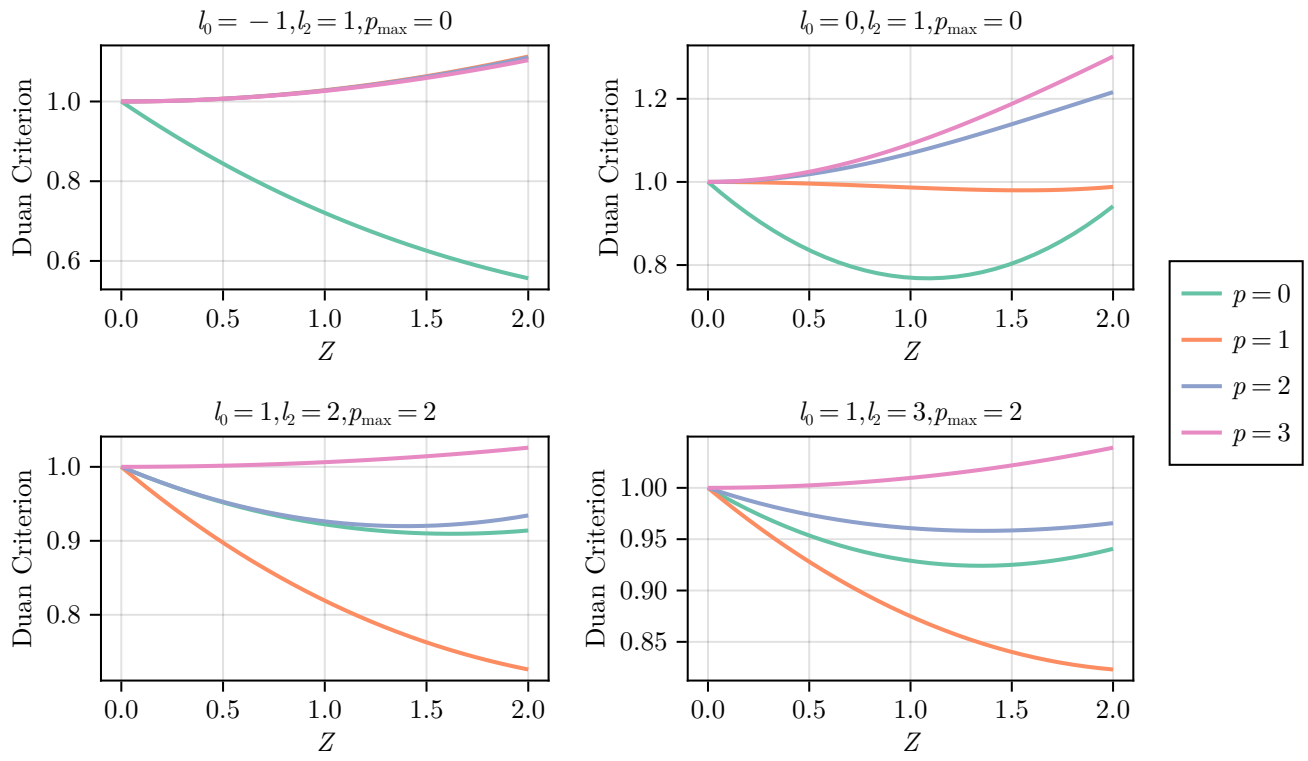


FIG. 1. Phase-optimized Duan quantity for the diffraction-free Bogoliubov solution. Values below one certify two-mode entanglement.

The crossing-pair argument explains why the first six cuts admit linear violations, but does not predict the optimized witness values or their near-degeneracies. The small violation for $\mathcal{A}\mathcal{B}_0\mathcal{B}_1|\mathcal{C}$ illustrates that modes outside the first-order radial support can nevertheless participate in entanglement at finite distance.

D. Genuine multipartite entanglement of selected modes

We now trace out the complement and ask whether the selected modes themselves are genuinely multipartite entangled. Their reduced state is generally mixed, even when the complete field is pure. Entanglement across every fixed bipartition is then insufficient: a biseparable state can have the form

$$\rho_{\text{bisep}} = \sum_{\lambda} p_\lambda \rho_{S_\lambda|T_\lambda}, \tag{48}$$

where each component is separable across a cut that can vary with λ. For example, a mixture of states separable across $\mathcal{A}\mathcal{B}_0|\mathcal{B}_1$, $\mathcal{A}\mathcal{B}_1|\mathcal{B}_0$, and $\mathcal{A}|\mathcal{B}_0\mathcal{B}_1$ need not be separable across any one fixed cut.

The MultiWit routine of Ref. [41] optimizes a covariance witness valid for the entire class of biseparable states. A value below one therefore certifies genuine multipartite entanglement of the selected modes, addressing the question left open by the grouped pure-state argument of Sec. IV C.

In Fig. 4, we show the witness for sets of the form $\{LG_{0,1}(w_2 = w)\} \cup \{LG_{p,3}(w_1) \mid p \le P\}$, with P varying from 0 to 3, while maintaining $l_0 = 2$. The $P = 0$ case is bipartite; for $P \ge 1$, a violation certifies genuine multipartite entanglement. The numerical results display a pattern related to the first-order radial support: the sets with $P \le p_{\max} = 1$ have lower witness values and hence larger margins below the separability threshold.

In particular, for $P = 1$ the analytical argument certifies the pure-state grouping $\mathcal{A}|\mathcal{B}_0|(\mathcal{B}_1\mathcal{R})$, whereas a MultiWit violation certifies the reduced three-mode state on $\mathcal{A}|\mathcal{B}_0|\mathcal{B}_1$. The latter shows that genuine multipartite entanglement survives discarding the remaining field modes. Violations for larger selected sets, when present, involve modes outside the first-order support and are finite-distance results within the Bogoliubov approximation.

VI. CONCLUSION

We studied bi- and multipartite entanglement properties in the radial structure of the modes generated by

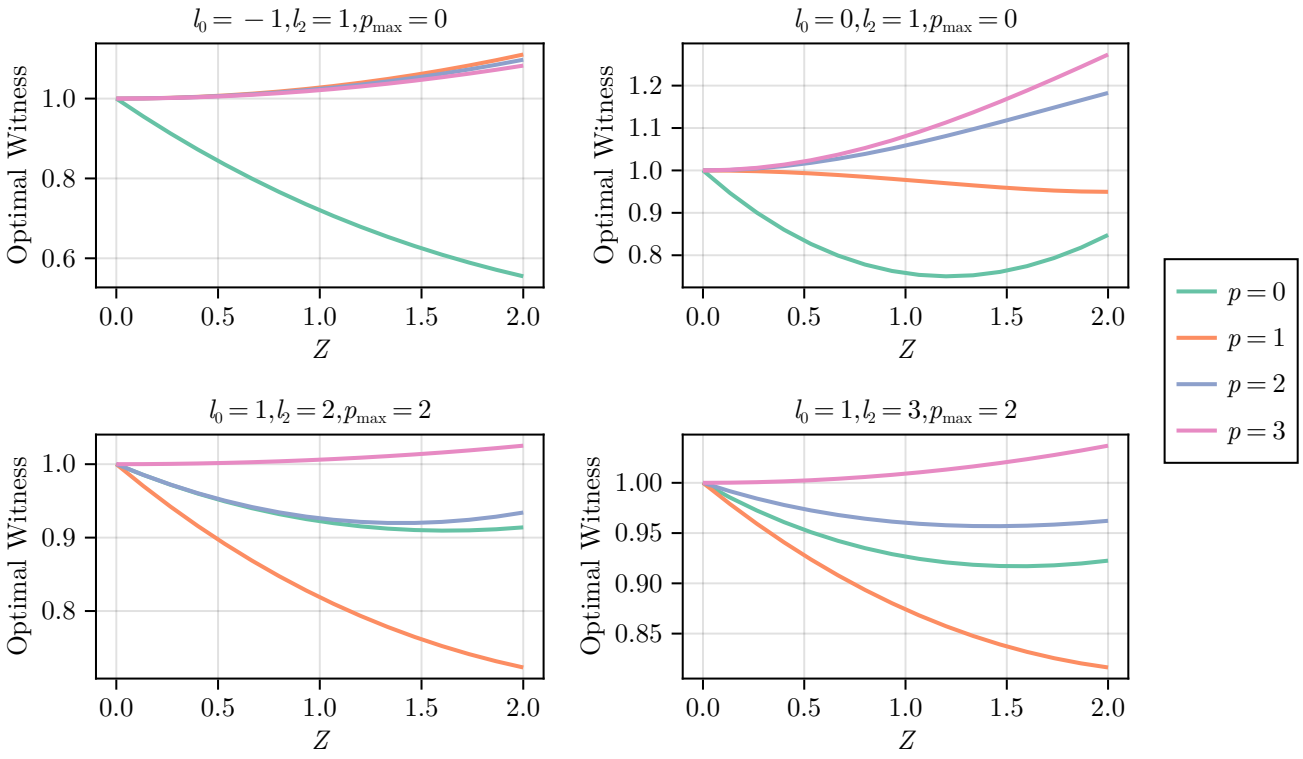


FIG. 2. Optimized witness value in the bipartite case.

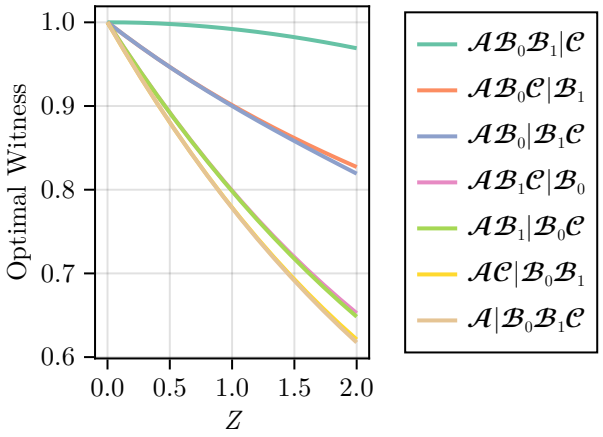


FIG. 3. Optimized witness value for bipartite entanglement over multiple modes with $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$.

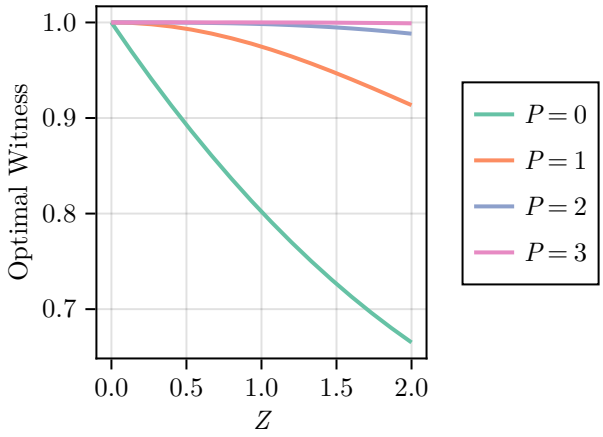


FIG. 4. Optimized witness value for genuine multipartite entanglement with $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2|l_0|, |l_2|) = 1$.

self-phase modulation with orbital angular momentum beams. The evolution of the mode structure in the nonlinear interaction is governed by chiral first-order selection rules, which constrain the radial support of the mean field and anomalous correlations. Moreover, we show that the radial structure of other OAM components can exhibit bi- and multipartite entanglement, depending on the relative chirality of the interacting modes. Pairwise violations establish entanglement across every bipartition of the coupled set. For the full pure field, including the remaining modes in one selected party yields a genuine multipartite certificate whose party count is controlled by

the OAM values at the specified detection waists. The first-order results are complemented by finite-distance calculations using the diffraction-free Bogoliubov solution, within the regime of negligible diffraction and small fluctuation backreaction. Optimized witnesses further certify entanglement in reduced sets of detection modes, including genuine multipartite entanglement after the remaining field is discarded. Our results can be investigated by current four-wave mixing and self-phase modulation setups in the softly focused regime, where the Rayleigh distance is much larger than the nonlinear interaction distance. This effect can be useful for building new quantum information platforms, where bi- and multipartite entanglement are controlled by orbital angular momentum.

Appendix A: Diffraction-free Bogoliubov evolution

The first-order calculation in the main text retains diffraction and all nonlinear fluctuation terms in Eq. (25). Here we start from the same full fluctuation equation and adopt a different approximation: we neglect diffraction over the interaction region and discard the quadratic and cubic fluctuation terms Ω of Eq. (26), while retaining the full dependence on propagation distance. The resulting Bogoliubov equations are solved without expanding in the propagation distance or the accumulated classical nonlinear phase.

With the same decomposition $\mathcal{E} = \alpha_c + \eta$ and input $\alpha(\mathbf{r}) = \alpha_c(\mathbf{r}, 0)$, these equations are

$$\begin{aligned} 2ik\partial_z\alpha_c &= g|\alpha_c|^2\alpha_c, \\ 2ik\partial_z\eta &= g\left(2|\alpha_c|^2\eta + \alpha_c^2\eta^\dagger\right). \end{aligned} \tag{A1}$$

The classical intensity is constant along propagation. Defining

$$I(\mathbf{r}) = |\alpha(\mathbf{r})|^2, \qquad \theta(\mathbf{r}, z) = \frac{gz}{2k}I(\mathbf{r}), \tag{A2}$$

we have

$$\alpha_c(\mathbf{r}, z) = \alpha(\mathbf{r})e^{-i\theta(\mathbf{r},z)}. \tag{A3}$$

1. Local fluctuation solution

Each transverse position evolves independently. Introduce the rotating-frame fluctuation $b(\mathbf{r}, z) = e^{i\theta(\mathbf{r},z)}\eta(\mathbf{r}, z)$. Suppressing the position argument temporarily, Eq. (A1) becomes

$$2ik\partial_z b = g\left(Ib + \alpha^2 b^\dagger\right). \tag{A4}$$

Together with its conjugate, this gives

$$2ik\partial_z \begin{pmatrix} b \\ b^\dagger \end{pmatrix} = gM \begin{pmatrix} b \\ b^\dagger \end{pmatrix}, \qquad M = \begin{pmatrix} I & \alpha^2 \\ -\alpha^{*2} & -I \end{pmatrix}. \tag{A5}$$

The matrix is independent of z and satisfies $M^2 = 0$. Therefore its exponential terminates exactly:

$$\exp\left(-\frac{igz}{2k}M\right) = \mathbb{I} - \frac{igz}{2k}M. \tag{A6}$$

Returning to η and writing $\eta_0(\mathbf{r}) = \eta(\mathbf{r}, 0)$, we obtain

$$\begin{aligned} \eta(\mathbf{r}, z) = e^{-i\theta}\Big[&(1 - i\theta)\eta_0(\mathbf{r}) \\ &- i\frac{gz}{2k}\alpha^2(\mathbf{r})\eta_0^\dagger(\mathbf{r})\Big]. \end{aligned} \tag{A7}$$

Although the rotating-frame coefficients are linear in z, this is not a first-order Taylor approximation. It is the exact solution of the diffraction-free linearized fluctuation equations.

Writing $\eta = A\eta_0 + B\eta_0^\dagger$, the coefficients are

$$A = e^{-i\theta}(1 - i\theta), \qquad B = -i\frac{gz}{2k}\alpha^2 e^{-i\theta}. \tag{A8}$$

They obey $|A|^2 - |B|^2 = 1$, so the field commutator is preserved. For the coherent input, $\langle\eta_0\rangle = 0$ and hence $\langle\eta(z)\rangle = 0$. Thus the mean field equals α_c within this approximation.

2. Field and mode correlations

The initial fluctuations satisfy

$$\begin{aligned} \langle\eta_0(\mathbf{r}_1)\eta_0^\dagger(\mathbf{r}_2)\rangle &= \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ \langle\eta_0^\dagger(\mathbf{r}_2)\eta_0(\mathbf{r}_1)\rangle &= \langle\eta_0(\mathbf{r}_1)\eta_0(\mathbf{r}_2)\rangle = 0. \end{aligned} \tag{A9}$$

Consequently the correlations are

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) &= A(\mathbf{r}_1)B(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ H(\mathbf{r}_1, \mathbf{r}_2; z) &= B(\mathbf{r}_1)B^*(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \end{aligned} \tag{A10}$$

Substituting the coefficients gives

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) = &\left[-\frac{igz}{2k} - \left(\frac{gz}{2k}\right)^2 I(\mathbf{r}_1)\right] \\ &\times \alpha_c^2(\mathbf{r}_1, z)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ H(\mathbf{r}_1, \mathbf{r}_2; z) = &\left(\frac{gz}{2k}\right)^2 I^2(\mathbf{r}_1)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \end{aligned} \tag{A11}$$

Using the main-text scaling, $\theta = Z|u|^2$ and $u_Z = u\exp\left(-iZ|u|^2\right)$, projection onto the detection modes yields

$$\begin{aligned} f(v_1, v_2) = &-iZ(v_1, v_2; u_Z, u_Z) \\ &- Z^2(v_1, v_2, u_Z; u_Z, u_Z, u_Z), \\ h(v_1, v_2) = &\, Z^2(v_1, u_Z, u_Z; v_2, u_Z, u_Z). \end{aligned} \tag{A12}$$

The phases in h cancel, whereas those in f retain the nonlinear classical evolution. Expanding Eq. (A12) at small

Z recovers $f = -iZ(v_1, v_2; u, u) + \mathcal{O}(Z^2)$ and $h = \mathcal{O}(Z^2)$, in agreement with the first-order calculation in the main text. At finite distance, these expressions are exact only within the diffraction-free Bogoliubov approximation, not for the full quantum Kerr equation.

The approximation requires negligible diffraction of the relevant transverse structures over the interaction length, as well as sufficiently small fluctuation interactions and backreaction on the classical field. A large pump Rayleigh range is a useful initial scale, but finer structure generated by the nonlinear phase can have shorter diffraction lengths. The derivation applies to any coherent input profile, including a coherent pump plus seeds.

The Bogoliubov approximation also permits retaining diffraction in both the classical background and the linear fluctuation equations. Quantum correlations can then be computed numerically by propagating the corresponding Green functions [42, 43], or by back-propagating the detection functions with the adjoint fluctuation equations [44, 45] and evaluating their correlations in the known input state. Such a numerical treatment with diffraction lies beyond the scope of the present work, which uses the analytical diffraction-free solution above.

Appendix B: Coherent seeding and first-order selection rules

1. Role of the seed in quadrature detection

For the single coherent pump $\alpha = A_p v_p$, with $v_p = LG_{0,l_p}(w)$, both terms in Eq. (B3) with $\epsilon = 0$ vanish for a detection mode of definite OAM $l_v \neq l_p$: the nonlinear mean-field source $|v_p|^2 v_p$ carries the same OAM as the pump. Thus, within this first-order treatment, self-phase modulation changes the mean field only in the $l = l_p$ sector, including its radial structure.

The two-mode correlations studied in the main text instead involve $l_1 + l_2 = 2l_p$ with $l_2 \neq l_p$, so neither detected mode has a coherent mean at this order. This does not prevent quadrature measurements with an external local oscillator, but a coherent amplitude in a detected mode can assist spatial alignment and phase control. The purpose of the seed is to supply such a reference while retaining, as closely as possible, the original entanglement correlations. Zero coherent mean should not be identified with exactly zero photon occupation: the centered population $h(v, v)$ starts beyond first order in distance.

We consider mutually coherent, collinear pump and seed fields sharing the frequency, polarization, and temporal mode implicit in our scalar model. We first derive the first-order correlations for an arbitrary spatial seed, and then examine seeding only v_2 or directly seeding both detected OAM families. Throughout this appendix, "first order" in propagation refers to the coefficient linear in distance; terms linear and quadratic in seed amplitude are distinguished explicitly.

2. First-order corrections for an arbitrary coherent seed

Let the input profile be

$$\alpha(\mathbf{r}, 0) = A_p \big[v_p(\mathbf{r}) + \epsilon s(\mathbf{r})\big], \tag{B1}$$

where $\|v_p\| = \|s\| = 1$ and the complex parameter ϵ specifies the overall seed amplitude and phase relative to the pump. The normalized profile s specifies its spatial composition; for example, $s = \sum_j c_j v_{s,j}$ in an orthonormal seed basis, with $\sum_j |c_j|^2 = 1$. We use the pump waist and occupation $N_p = A_p^2$ to define

$$\begin{aligned} U_p &= w v_p, \\ S &= w s, \\ U &= U_p + \epsilon S, \\ Z_p &= \frac{g z N_p}{2kw^2}. \end{aligned} \tag{B2}$$

The profiles U_p, S, U are dimensionless, and Z_p is held fixed when comparing different seeds. Weak seeding means $|\epsilon| \ll 1$; it need not imply a seed occupation smaller than one. If the seed is orthogonal to the pump, its occupation relative to the pump is $q = |\epsilon|^2$, and the main-text normalization is $u = U/\sqrt{1+q}$, $Z = (1+q)Z_p$. More generally, replace $1 + q$ by $\|v_p + \epsilon s\|^2$. These conventions give the same product $Zu^2 = Z_p U^2$.

The input remains a coherent state of a spatial field distribution. Therefore, the mean-field formula and the general correlation derivation apply directly to the combined profile. For example, in freely propagated detection modes, with the nonlinear overlap evaluated at the input plane,

$$\langle a(v, z)\rangle = \frac{A_p}{w}\left[(v, U) - iZ_p(v, U; U, U)\right] + O(Z_p^2). \tag{B3}$$

For orthonormal detection modes v_1, v_2, define

$$\begin{aligned} C_{12} &= (v_1, v_2; U_p, U_p), \\ \delta C_{12}^{(1)} &= 2\epsilon(v_1, v_2; U_p, S), \\ \delta C_{12}^{(2)} &= \epsilon^2(v_1, v_2; S, S). \end{aligned} \tag{B4}$$

The superscripts indicate order in seed amplitude, not distance. Expanding $U^2 = (U_p + \epsilon S)^2$ gives

$$\begin{aligned} f(v_1, v_2) &= -iZ_p\left[C_{12} + \delta C_{12}^{(1)} + \delta C_{12}^{(2)}\right] + O(Z_p^2), \\ h(v_1, v_2) &= O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_p|\left|C_{12} + \delta C_{12}^{(1)} + \delta C_{12}^{(2)}\right| + O(Z_p^2). \end{aligned} \tag{B5}$$

This is the complete coefficient linear in distance for an arbitrary coherent seed. It contains no terms beyond quadratic order in seed amplitude. Because f and h are centered correlations, the seed's coherent displacement

alone does not contribute to them; the corrections above arise from its participation in the nonlinear interaction.

The angular content determines which corrections survive. Write

$$U_p = R_p(r)e^{il_p\phi}, \qquad S = \sum_m S_m(r)e^{im\phi}. \tag{B6}$$

For detection OAM sum $L = l_1 + l_2$, the pump–seed term requires $l_p + m = L$, and the seed–seed term requires $m + n = L$. In the original channel $L = 2l_p$, the relevant harmonic is

$$[U^2]_{2l_p} = e^{2il_p\phi}\left[R_p^2 + 2\epsilon R_p S_{l_p} + \epsilon^2 \sum_m S_m S_{2l_p-m}\right]. \tag{B7}$$

Thus only a seed component with pump OAM can modify this channel linearly in seed amplitude. Quadratic corrections require complementary seed OAM values summing to $2l_p$. The radial overlaps must also be nonzero. In particular, a seed with pump OAM but a different radial profile can alter the radial rule at linear order in seed amplitude; a seed proportional to v_p simply changes the combined pump amplitude.

3. Seeding only the mode v_2

Consider the same detection family as in the main text,

$$\begin{aligned} v_p &= LG_{0,l_p}(w), \\ v_2 &= LG_{0,l_2}(w_2), \\ v_{1,p} &= LG_{p,l_1}(w_1), \\ l_1 &= 2l_p - l_2, \\ \frac{1}{w_1^2} &= \frac{2}{w^2} + \frac{1}{w_2^2}, \qquad l_2 \neq l_p. \end{aligned} \tag{B8}$$

Here l_p denotes pump OAM, whereas the index p of $v_{1,p}$ denotes radial order. Set $s = v_2$, so $S = U_2$, where $U_2 = wv_2$. The pump–seed and seed–seed terms carry OAM $l_p + l_2$ and $2l_2$, respectively. Neither equals $2l_p$, so both corrections vanish in the original channel. Defining $C_p = (v_{1,p}, v_2; U_p, U_p)$ gives

$$\begin{aligned} f(v_{1,p}, v_2) &= -iZ_pC_p + O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_pC_p| + O(Z_p^2). \end{aligned} \tag{B9}$$

The coefficient linear in distance is independent of ϵ, including its phase. Hence the original cutoff

$$p_{\max} = |l_p| + \frac{|l_2| - |2l_p - l_2|}{2} \tag{B10}$$

is preserved exactly at this order for the detection profiles in (B8): $C_p = 0$ for $p > p_{\max}$. The seed does open other OAM channels, at orders $Z_p\epsilon$ and $Z_p\epsilon^2$, but these do not enter the selected pair's first-order coefficient.

The mean-field benefit is twofold. The seeded mode has the input amplitude

$$\langle a(v_2, 0)\rangle = A_p\epsilon. \tag{B11}$$

Moreover, the nonlinear source $|U|^2U$ contains $\epsilon^* U_p^2 U_2^*$, whose OAM is $2l_p - l_2 = l_1$. Projection of (B3) therefore yields

$$\langle a(v_{1,p}, z)\rangle = -iZ_pA_p\epsilon^*C_p + O(Z_p^2). \tag{B12}$$

A seed in v_2 can thus generate coherent references in its conjugate radial modes while preserving the original first-order correlations. These stimulated amplitudes are suppressed by Z_p, follow the radial coefficients C_p, and vanish at this order outside the cutoff. Whether they suffice for homodyne phase control depends on the experimental sensitivity.

4. Seeding both detected OAM families

To supply a direct coherent reference in every selected detection mode, including radial modes whose stimulated amplitudes are too small, take

$$S = c_2U_2 + \sum_{p\in\mathcal{S}} c_{1,p}U_{1,p}, \tag{B13}$$

where $U_{1,p} = wv_{1,p}$ and $\mathcal{S}$ is the finite set of seeded radial orders. The two OAM families and the pump are mutually orthogonal, as are the radial modes within the first family. Normalization of s requires

$$|c_2|^2 + \sum_{p\in\mathcal{S}} |c_{1,p}|^2 = 1, \tag{B14}$$

the seed power fraction is $q = |\epsilon|^2$, and the input means are $\langle a(v_2, 0)\rangle = A_p\epsilon c_2$ and $\langle a(v_{1,p}, 0)\rangle = A_p\epsilon c_{1,p}$. These directly injected amplitudes are not suppressed by propagation distance.

Neither seed family has pump OAM, so $\delta C^{(1)}$ still vanishes in the original channel. However, their OAM values are complementary: $l_1 + l_2 = 2l_p$. Equation (B7) now gives

$$[U^2]_{2l_p} = U_p^2 + 2\epsilon^2 c_2 \sum_{p\in\mathcal{S}} c_{1,p}U_2U_{1,p}. \tag{B15}$$

For a detected radial order r, define

$$\begin{aligned} K_{rp} &= (v_{1,r}, v_2; U_2, U_{1,p}) \\ &= w^2 \int d^2\mathbf{r}\, v_{1,r}^* v_{1,p}|v_2|^2, \\ \delta C_r &= 2\epsilon^2 c_2 \sum_{p\in\mathcal{S}} c_{1,p}K_{rp}. \end{aligned} \tag{B16}$$

The first-order correlations become

$$\begin{aligned} f(v_{1,r}, v_2) &= -iZ_p(C_r + \delta C_r) + O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_p|\,|C_r + \delta C_r| + O(Z_p^2). \end{aligned} \tag{B17}$$

The correction is quadratic in seed amplitudes but remains linear in distance. Its phase relative to C_r depends on the products $\epsilon^2 c_2 c_{1,p}$, so complementary seeds can enhance or reduce the magnitude of an allowed correlation.

The weighted overlaps K_{rp} need not obey the original radial cutoff. Orthogonality of radial LG modes uses an unweighted inner product, whereas the factor $|v_2|^2$ changes the radial weight and Gaussian envelope. In particular,

$$K_{rr} = w^2 \int d^2\mathbf{r}\, |v_{1,r}|^2 |v_2|^2 > 0 \tag{B18}$$

for overlapping LG profiles. Thus directly seeding $v_{1,r}$ and v_2 can remove a first-order zero even when $r > p_{\max}$ and $C_r = 0$. Multiple radial seeds may interfere, with cancellation possible only for particular amplitudes and phases.

For a fixed normalized seed profile, $\delta C_r = O(\epsilon^2)$ and $q = |\epsilon|^2$. The original radial rule is therefore preserved through linear order in seed amplitude, but generally modified at quadratic order in seed amplitude, or first order in seed power fraction at fixed seed composition. For an allowed pair, the condition

$$|\delta C_r| \ll |C_r| \tag{B19}$$

ensures that its first-order correlation is approximately unchanged. For a forbidden pair, any nonzero δC_r lifts the first-order zero, however small its absolute magnitude.

Seeding only v_2 therefore preserves the original first-order radial rule exactly while relying on stimulated means in the complementary family. Seeding both families supplies independent input references in all selected modes, at the cost of quadratic seed corrections to that rule. These statements concern the first-order two-mode criterion; they do not establish a multipartite selection rule or a finite-distance witness violation. Beyond the small-distance regime, the combined input may be inserted directly into the correlation formulas of Appendix A, within their stated assumptions.

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